ECON 131 — Public Economics

Midterm 2 Practice Problem Set

Spring 2026 · Prof. Danny Yagan · UC Berkeley

Real Exam Format

Part I — True / False / Uncertain: 60 pts (Lectures 08–14)
Part II — Empirical Applications: 30 pts (DD Regression)
Part III — Analytic Problems: 60 pts (Lectures 07–11)
Total: 150 pts · 80 minutes

This practice set is intentionally more comprehensive than the real exam.
Answers are collected at the end of the document — do not peek until you've attempted everything.

PART I — TRUE / FALSE / UNCERTAIN

For each statement, state whether it is True, False, or Uncertain. Then explain your answer in 3–5 sentences. Full credit requires a clear, correct explanation — the verdict alone earns no points.

1. Evidence from the EITC shows that low-income workers respond more on the extensive margin (whether to work) than the intensive margin (how much to work), which is consistent with optimal tax theory's recommendation for wage subsidies at the bottom of the income distribution.

(Lecture 08, EITC labor supply — extensive vs. intensive margin)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

2. Saez's bunching evidence at kink points in the US income tax schedule shows that ordinary wage earners exhibit large real labor supply responses to marginal tax rates.

(Lecture 08, bunching evidence — real responses)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

3. Studies of top tax rate changes in Denmark show that high earners who moved to lower-tax municipalities represent real economic efficiency losses from taxation.

(Lecture 08, real vs. avoidance responses — migration)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

4. Feldstein's finding of large elasticities of taxable income after the 1986 Tax Reform Act implies that high tax rates cause substantial deadweight loss.

(Lecture 08, taxable income elasticity — Feldstein vs. Gruber-Saez)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

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5. In a difference-in-differences regression $Y_{it} = \alpha + \beta \cdot Treated_i + \gamma \cdot Post_t + \delta \cdot (Treated_i \times Post_t) + \varepsilon_{it}$, the coefficient γ captures the average difference in outcomes between the treatment and control groups before the policy change.

(Lecture 08B, DD regression — coefficient interpretation)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

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6. The key identifying assumption for difference-in-differences is that the treatment and control groups must have the same level of the outcome variable before the policy change.

(Lecture 08B, DD — parallel trends assumption)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

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7. Under the traditional view of dividend taxation, a cut in dividend tax rates will increase corporate investment because it lowers the cost of equity-financed investment.

(Lecture 09, traditional vs. new view of dividend taxation)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

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8. Under the new view of dividend taxation, the dividend tax reduces corporate investment by raising the cost of capital for firms that finance investment with retained earnings.

(Lecture 09, new view of dividend taxation)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

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9. If the government allows full expensing of investment (100% immediate deduction), then the effective tax rate on corporate investment is zero regardless of the statutory corporate tax rate.

(Lecture 09, cost of capital — full expensing)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

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10. Yagan (2015) found that the 2003 dividend tax cut led to a large increase in C-corporation investment relative to S-corporations.

(Lecture 09, Yagan 2015 — dividend tax cut and investment)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

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11. S-corporations are subject to double taxation because both the corporate entity and the individual shareholders pay income tax on profits.

(Lecture 09, C-corp vs. S-corp entity taxation)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

12. In the standard life-cycle model with no bequests, a tax on capital income unambiguously reduces the amount individuals save for retirement.

(Lecture 10, life-cycle model — savings response to capital taxation)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

13. A proportional consumption tax and a proportional labor income tax are equivalent in present value, assuming no initial wealth.

(Lecture 10, consumption tax equivalence)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

14. The Atkinson-Stiglitz theorem proves that capital income should never be taxed under any circumstances.

(Lecture 10, Atkinson-Stiglitz theorem — conditions and limits)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

15. Under the "accidental bequest" motive, estate taxes have no effect on the donor's savings behavior because bequests are unplanned.

(Lecture 10, bequest motives — accidental)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

16. A Pigouvian tax set equal to the marginal damage at the socially optimal quantity will correct a negative externality and achieve the efficient outcome.

(Lecture 11, corrective taxation — Pigouvian tax)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

17. The Coase theorem implies that government intervention is never needed to correct externalities, as private parties will always bargain to an efficient outcome.

(Lecture 11, Coase theorem — conditions and limits)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

18. When there is uncertainty about the marginal abatement cost curve and the marginal damage curve is relatively steep, a quantity instrument (cap-and-trade permits) is preferred to a price instrument (tax).

(Lecture 11, Weitzman — taxes vs. permits under uncertainty)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

19. If a factory's production creates pollution, the socially efficient level of pollution is always zero.

(Lecture 11, externalities — efficient pollution level)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

20. The Samuelson rule for efficient public goods provision states that each individual's marginal rate of substitution between the public good and private good should equal the marginal cost of the public good.

(Lecture 12, Samuelson rule — public goods optimality)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

21. Empirical evidence on charitable giving suggests that government grants to nonprofit organizations crowd out private donations approximately dollar-for-dollar, consistent with the theoretical prediction of a pure public goods model.

(Lecture 12, crowding out — theory vs. evidence)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

22. A public good that is non-rival but excludable (such as a streaming service) is called a "club good."

(Lecture 12, public goods classification)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

23. The Tiebout model shows that when residents can freely choose which jurisdiction to live in, competition among local governments leads to efficient provision of local public goods, similar to how competition among firms produces efficiency in private markets.

(Lecture 14, Tiebout model — voting with feet)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

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24. Property tax capitalization means that an increase in the local property tax rate will be fully offset by a decrease in property values, leaving homebuyers indifferent between high-tax and low-tax communities.

(Lecture 14, property tax capitalization)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

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25. California's Proposition 13, which caps property tax increases for existing homeowners, encourages housing mobility because residents face lower taxes and can afford to move more easily.

(Lecture 14, Proposition 13 — lock-in effect)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

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26. The flypaper effect describes the empirical finding that lump-sum intergovernmental grants lead to higher local government spending than equivalent increases in residents' private income, contrary to standard economic theory.

(Lecture 14, flypaper effect — grants and local spending)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

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27. Matching grants and block grants have the same effect on local government spending because both increase the resources available to the local government.

(Lecture 14, intergovernmental grants — matching vs. block)

Verdict: ☐ TRUE ☐ FALSE ☐ UNCERTAIN

Explanation:

PART II — EMPIRICAL APPLICATIONS

Answer all parts. Show your work clearly.

Question E1: NOx Budget Program and Housing Values

In 2003, the EPA implemented the NOx Budget Program (NBP), requiring large pollution sources in certain eastern US states to reduce nitrogen oxide emissions. Researchers want to estimate the effect of this pollution reduction on local housing values.

They collect data on average housing values in counties near regulated power plants ("treatment" counties) and counties far from any regulated plant ("control" counties), for years 2001 (before) and 2005 (after).

Table: Average Housing Values ($1,000s)

	Before (2001)	After (2005)
Treatment counties	$180	$220
Control counties	$200	$225

(a) Write the difference-in-differences regression equation that the researchers would estimate, defining all variables clearly.

(Lecture 08B, DD regression equation setup)

(b) Using the table above, calculate the DD estimate of the effect of the NBP on housing values. Show each step.

(Lecture 08B, DD calculation from 2×2 table)

(c) In the regression $Y_{it} = \alpha + \beta \cdot Treated_i + \gamma \cdot Post_t + \delta \cdot (Treated_i \times Post_t) + \varepsilon_{it}$, what are the numerical values of α, β, γ, and δ based on the table?

(Lecture 08B, DD coefficient interpretation)

(d) State the key identifying assumption for this DD design. Describe a scenario in which this assumption would be violated.

(Lecture 08B, parallel trends assumption)

(e) Suppose you observe that housing values in treatment counties were growing faster than control counties even before 2003. How would this affect your interpretation of the DD estimate?

(Lecture 08B, parallel trends violation)

Question E2: Tax Reform and Earnings Responses

A country implements a tax reform in 2010 that sharply reduces the top marginal tax rate for individuals earning above $200,000 in Region A, but does not change rates in Region B. Researchers collect data on average reported taxable income among high earners in both regions:

	Before (2009)	After (2011)
Region A (tax cut)	$250,000	$285,000
Region B (no change)	$245,000	$260,000

(a) Calculate the DD estimate of the effect of the tax cut on reported taxable income.

(Lecture 08B, DD calculation)

(b) Can we conclude from this DD estimate that the tax cut caused a real increase in labor supply? Why or why not?

(Lecture 08, real vs. avoidance responses)

(c) Suggest one type of avoidance response and one type of real response that could explain the increase in reported income in Region A.

(Lecture 08, real vs. avoidance responses — examples)

PART III — ANALYTIC PROBLEMS

Show all work. Partial credit is available for correct intermediate steps.

Question A1: Optimal Top Income Tax Rate

Suppose the government is considering raising the top marginal tax rate on income above \$500,000. Currently the top marginal rate is $\tau = 0.40$. Assume:

- The elasticity of taxable income for top earners is $e = 0.25$
- The Pareto parameter of the income distribution is $a = 2$
- The average income of those above \$500K is \$1,000,000
- There are 1,000 taxpayers above the \$500K threshold

(a) Using the formula $\tau^* = 1/(1 + a \cdot e)$, calculate the revenue-maximizing top marginal tax rate. Is the current rate above or below this?

(Lecture 07, optimal top rate formula)

(b) Calculate the mechanical revenue gain from raising the top rate by $\Delta\tau = 0.05$ (from 0.40 to 0.45). Use: $\Delta M = \Delta\tau \times (\bar{z} - \bar{z}^*) \times N$, where $\bar{z} = \$1{,}000{,}000$ is the average income of top earners, $\bar{z}^* = \$500{,}000$ is the threshold, and $N = 1{,}000$.

(Lecture 07, Laffer curve — mechanical effect)

(c) Calculate the behavioral revenue loss from this rate increase. Use: $\Delta B = -\tau \times e \times [\Delta\tau/(1-\tau)] \times \bar{z} \times N$, where $\tau = 0.40$.

(Lecture 07, Laffer curve — behavioral effect)

(d) What is the net revenue effect of the rate increase? Should the government raise the rate based on a revenue-maximizing criterion?

(Lecture 07, Laffer curve — net revenue)

(e) Suppose a new study estimates $e = 0.8$ instead of 0.25. Recalculate τ^* and explain how this changes your policy recommendation.

(Lecture 07, optimal top rate — elasticity sensitivity)

Question A2: Corporate Tax and Cost of Capital

A firm is considering purchasing a machine that costs \$100,000 and depreciates over 5 years (straight-line). The real interest rate is $r = 5\%$. The corporate tax rate is $\tau = 30\%$.

(a) With no taxes, what is the minimum annual pre-tax return the machine must produce for the firm to invest (i.e., what is the cost of capital)? Express as a rate.

(Lecture 09, cost of capital — no-tax baseline)

(b) Now suppose the firm uses straight-line depreciation (deducts \$20,000 per year for 5 years). Calculate the present value of depreciation deductions per dollar invested, using $PV = \Sigma(\tau \cdot d_t)/(1+r)^t$, where $d_t = 0.20$ for $t = 1$ to 5. (Hint: this is just τ times the PV of a 5-year annuity of \$0.20.)

(Lecture 09, cost of capital — depreciation PV)

(c) Using the Hall-Jorgenson formula: cost of capital = $(r + \delta)(1 - \tau \cdot z)/(1 - \tau)$, where $\delta = 0.20$ (depreciation rate) and z is the PV of depreciation deductions per dollar of investment from part (b) divided by τ. Calculate the cost of capital.

(Lecture 09, Hall-Jorgenson cost-of-capital formula)

(d) Now suppose Congress allows full expensing ($z = 1$). Recalculate the cost of capital. Explain intuitively why this result occurs.

(Lecture 09, full expensing — cost of capital)

(e) Under which depreciation regime (straight-line or full expensing) would a cut in the corporate tax rate τ increase investment more? Explain.

(Lecture 09, tax policy — depreciation and investment)

Question A3: Pollution Externality

A competitive steel industry produces output Q. The industry's private marginal cost and marginal damage from pollution are:

$PMC(Q) = 20 + 2Q$

$MD(Q) = Q$ (marginal damage from pollution)

Demand (marginal benefit): $P = 140 - 2Q$

(a) Find the market equilibrium quantity Q_m and price P_m (where Demand = PMC).

(Lecture 11, externalities — market equilibrium)

(b) Write the expression for Social Marginal Cost (SMC = PMC + MD). Find the socially optimal quantity Q^* and price P^*.

(Lecture 11, externalities — social optimum)

(c) Calculate the deadweight loss from the externality (the area of the triangle between SMC and Demand, from Q^* to Q_m).

(Lecture 11, externalities — deadweight loss)

(d) What Pigouvian tax per unit of output would achieve the social optimum? Verify that the market produces Q* under this tax.

(Lecture 11, Pigouvian tax — calculation)

(e) Suppose the government is uncertain whether the true MD is Q or 3Q. The true PMC is known. Compare the welfare consequences of (i) setting a tax based on the expected MD = 2Q vs. (ii) issuing Q* permits. Which instrument is better if damages are actually 3Q? Explain using Weitzman's logic.

(Lecture 11, Weitzman — taxes vs. permits under uncertainty)

Question A4: Taxation of Savings

Consider a two-period life-cycle model. An individual earns W = $100,000 in period 1 (working period) and nothing in period 2 (retirement). The interest rate is r = 10%. The individual's utility function is $U = \ln(C_1) + \ln(C_2)$.

(a) With no taxes, solve for optimal C_1, C_2, and savings S. (Hint: set up the intertemporal budget constraint and use the first-order condition MRS = 1+r.)

(Lecture 10, life-cycle model — no-tax optimum)

(b) Now suppose the government imposes a 50% tax on capital income (interest). The after-tax interest rate becomes $r(1-t) = 0.10(0.50) = 0.05$. Solve for the new C_1, C_2, and savings S.

(Lecture 10, capital income tax — life-cycle model)

(c) Did savings increase or decrease? Decompose the change into substitution and income effects and explain intuitively which dominates for this utility function.

(Lecture 10, substitution vs. income effects on savings)

(d) Now instead of taxing capital income, suppose the government imposes a 20% proportional tax on labor income (W). Solve for optimal C_1, C_2, and S with this tax. Compare the present value of total tax revenue collected under this scheme versus part (b).

(Lecture 10, labor income tax vs. capital income tax)

ANSWER KEY

Do not read until you have attempted all questions.

Part I — True / False / Uncertain Answers

1. TRUE. Empirical EITC studies find large participation responses but small hours-of-work responses. This supports the optimal tax argument for phasing in subsidies to draw people into the labor force (extensive margin), which is where the real response occurs.

2. FALSE. FALSE. Saez finds very little bunching at kink points for most US wage earners, suggesting small intensive-margin responses. The bunching that does appear is concentrated among the self-employed and likely reflects reporting/avoidance behavior, not real labor supply changes.

3. FALSE. FALSE. Geographic migration in response to taxes is an avoidance response, not a real response. The earner's productive output doesn't change — they simply relocate to a lower-tax jurisdiction. Only real responses (working fewer hours, less effort) create deadweight loss.

4. UNCERTAIN. UNCERTAIN. Feldstein found large elasticities (~1.0), but Gruber and Saez showed these estimates were biased upward by mean reversion and income trends. After correcting, the elasticity drops to ~0.4. So whether the deadweight loss is "substantial" depends on which estimate you trust. The corrected estimates suggest moderate, not enormous, deadweight loss.

5. FALSE. FALSE. γ captures the change in the outcome between pre and post periods that is common to both groups (the time trend). β captures the pre-existing level difference between treatment and control groups.

6. FALSE. FALSE. DD does NOT require equal levels — it requires parallel trends. The groups can start at different levels, but absent treatment, they must have followed the same trend over time. β captures any level difference.

7. TRUE. TRUE. Under the traditional view, firms finance marginal investment with new equity. Dividends are the return to new shareholders, so taxing dividends raises the required pre-tax return, increasing cost of capital. Cutting dividend taxes lowers this cost, stimulating investment.

8. FALSE. FALSE. Under the new view, firms finance marginal investment from retained earnings. The dividend tax applies symmetrically to the alternative (paying dividends now) and the return (paying dividends later), so it cancels out and does not affect the cost of capital or investment decisions.

9. TRUE. TRUE. With full expensing, the present value of depreciation deductions equals the full cost of the investment. In the Hall-Jorgenson formula, this makes the cost of capital independent of the tax rate. Intuitively, the government is now a silent partner: it pays τ share of the cost and receives τ share of the return.

10. FALSE. FALSE. Yagan used S-corps as a control group for C-corps and found no significant increase in C-corp investment after the 2003 dividend tax cut. This is evidence for the new view of dividends (tax doesn't affect investment), not the traditional view.

11. FALSE. FALSE. S-corps are pass-through entities: profits flow directly to shareholders' personal tax returns and are taxed only once at the individual level. C-corps face double taxation (corporate tax on profits + personal tax on dividends).

12. FALSE. FALSE. The effect is theoretically ambiguous. The substitution effect discourages saving (lower after-tax return), but the income effect encourages more saving (you need to save more to reach the same retirement consumption target). The net effect depends on which dominates.

13. TRUE. TRUE. The lifetime budget constraint shows that PV(consumption) = PV(labor income) when there is no initial wealth. A tax on one side is equivalent to a tax on the other. Both exempt the normal return to saving from taxation.

14. FALSE. FALSE. The theorem says that with an optimal nonlinear income tax, capital income taxation is unnecessary IF preferences over consumption and leisure are separable (utility is weakly separable). When this fails — e.g., rich people have higher savings rates independent of earnings — capital income taxation can improve welfare.

15. TRUE. TRUE. If bequests are accidental (people save for precautionary/retirement reasons and happen to die with wealth left over), then the estate tax doesn't affect savings decisions because people weren't saving in order to leave a bequest.

16. TRUE. TRUE. A Pigouvian tax of $t = MD(Q^*)$ shifts the private cost curve up to equal the social marginal cost at the optimum, inducing producers to internalize the externality and produce the socially efficient quantity.

17. FALSE. FALSE. Coase theorem requires well-defined property rights, zero transaction costs, and complete information. In practice, many externalities (e.g., air pollution affecting millions) involve high transaction costs, making private bargaining infeasible. Government intervention is often necessary.

18. TRUE. TRUE. Weitzman (1974) showed that when marginal damages are steep (damages accelerate quickly with additional pollution), getting the quantity right is critical. Permits fix the quantity, avoiding catastrophically high damages. A tax might allow too much pollution if costs are lower than expected.

19. FALSE. FALSE. The efficient level of pollution is where SMC = SMB (or equivalently, marginal abatement cost = marginal damage). Reducing pollution to zero would require shutting down valuable production. Efficiency means balancing the cost of pollution against the benefit of the goods produced.

20. FALSE. FALSE. The Samuelson rule states that the SUM of all individuals' MRS must equal the marginal cost: $\Sigma MRS_i = MC$. Because public goods are non-rival, everyone consumes the same unit, so we add up willingness to pay across all individuals.

21. FALSE. FALSE. The pure public goods model predicts complete (dollar-for-dollar) crowding out, but empirical evidence finds only partial crowding out — typically around $0.10-$0.30 per dollar of government grants. This is because people give for reasons beyond the public good itself (warm glow).

22. TRUE. TRUE. Goods are classified by rivalry and excludability. Non-rival + non-excludable = pure public good. Non-rival + excludable = club good. Rival + non-excludable = common resource. A streaming service is non-rival (one person watching doesn't reduce others' ability to watch) but excludable (subscription required).

23. TRUE. TRUE. Tiebout's "voting with feet" mechanism suggests that mobile residents sort into communities matching their preferences for public goods and taxes, creating competitive pressure for efficient provision. However, this requires strong assumptions (full mobility, many jurisdictions, perfect information, no spillovers).

24. TRUE. TRUE. If property taxes are capitalized, a higher tax reduces the asset value of the home by the present value of future extra taxes. A buyer is indifferent: a cheap house in a high-tax area costs the same total (price + PV of taxes) as an expensive house in a low-tax area, assuming equal public services.

25. FALSE. FALSE. Prop 13 creates a lock-in effect that discourages mobility. When you sell and buy a new home, the new property is reassessed at market value, causing a tax jump. This makes longtime owners reluctant to move, reducing housing turnover.

26. TRUE. TRUE. Standard theory says a lump-sum grant should be equivalent to increasing residents' income — most should be "returned" via tax cuts. But empirically, money "sticks where it hits": grants to governments get spent on public services rather than rebated as tax cuts.

27. FALSE. FALSE. Matching grants have both income and substitution effects (they lower the relative price of public goods), while block grants have only an income effect. Matching grants therefore stimulate more local spending on the matched service. They are not equivalent.

Part II — Empirical Answers

E1:

(a) $Y_{it} = \alpha + \beta \cdot Treated_i + \gamma \cdot Post_t + \delta \cdot (Treated_i \times Post_t) + \varepsilon_{it}$, where Y_{it} is housing value in county i at time t, $Treated_i = 1$ for counties near regulated plants, $Post_t = 1$ for year 2005.

(b) DD = (220 - 180) - (225 - 200) = 40 - 25 = $15,000. Treatment counties gained $40K, control gained $25K, so the NBP caused an extra $15K in housing value.

(c) $\alpha = 200$ (control, before). $\beta = 180 - 200 = -20$ (treatment is $20K lower than control before policy). $\gamma = 225 - 200 = 25$ (common time change). $\delta = 15$ (the DD estimate).

(d) The key assumption is parallel trends: absent the NBP, housing values in treatment and control counties would have followed the same trend. Violation example: treatment counties were near growing cities and would have appreciated faster anyway, even without the clean air regulation.

(e) If treatment counties were already growing faster pre-2003, the parallel trends assumption fails. The DD estimate of $15K would be biased upward — part of the estimated effect reflects the pre-existing differential trend, not the causal impact of the NBP.

E2:

(a) DD = (285,000 - 250,000) - (260,000 - 245,000) = 35,000 - 15,000 = $20,000. The tax cut is associated with a $20,000 increase in reported income.

(b) No. The increase in reported taxable income could reflect real responses (more hours worked, more effort) OR avoidance responses (income shifting, re-timing, reclassification). The elasticity of taxable income captures both. We cannot distinguish them from this DD alone.

(c) Avoidance example: high earners shift income from deferred compensation to current salary to take advantage of the lower rate (re-timing). Real example: professionals work longer hours or take on more clients because the after-tax reward per hour is higher.

Part III — Analytic Answers

A1: Optimal Top Income Tax Rate

(a) $\tau^* = 1/(1 + a \cdot e) = 1/(1 + 2 \times 0.25) = 1/1.5 = 0.667$ or 66.7%. Current rate 40% is below the revenue-maximizing rate, so there is room to raise rates and increase revenue.

(b) $\Delta M = 0.05 \times (1{,}000{,}000 - 500{,}000) \times 1{,}000 = 0.05 \times 500{,}000 \times 1{,}000 = \$25{,}000{,}000$ (mechanical gain of \$25M).

(c) $\Delta B = -0.40 \times 0.25 \times (0.05/0.60) \times 1{,}000{,}000 \times 1{,}000 = -0.40 \times 0.25 \times 0.0833 \times 1{,}000{,}000{,}000 = -\$8{,}333{,}333$ (behavioral loss of ~\$8.3M).

(d) Net = \$25M - \$8.3M = +\$16.7M. Positive, so the rate increase raises revenue. This is consistent with being below $\tau^* = 66.7\%$. The government should raise the rate.

(e) $\tau^* = 1/(1 + 2 \times 0.8) = 1/2.6 = 0.385$ or 38.5%. Now the current rate of 40% exceeds the revenue-maximizing rate. Raising it further would lose revenue. This shows the optimal rate is very sensitive to the elasticity estimate.

A2: Corporate Tax and Cost of Capital

(a) With no taxes, the cost of capital is simply $r + \delta = 0.05 + 0.20 = 0.25$ or 25%. The machine must earn at least 25% annually (5% opportunity cost + 20% depreciation replacement).

(b) PV of deductions per \$1 invested: $z = \Sigma (0.20)/(1.05)^t$ for $t=1$ to $5 = 0.20 \times [(1 - 1.05^{-5})/0.05] = 0.20 \times 4.3295 = 0.8659$. The PV of tax savings $= \tau \cdot z = 0.30 \times 0.8659 = 0.2598$.

(c) Cost of capital $= (r + \delta)(1 - \tau \cdot z)/(1 - \tau) = (0.05 + 0.20)(1 - 0.2598)/(1 - 0.30) = 0.25 \times 0.7402/0.70 = 0.25 \times 1.0574 = 0.2644$ or 26.4%. The tax raises the cost of capital above the no-tax level because depreciation deductions don't fully offset the tax.

(d) With $z = 1$: cost of capital $= (0.25)(1 - 0.30)/(1 - 0.30) = 0.25 \times 1.0 = 0.25 = 25\%$. Same as the no-tax case. With full expensing, the government effectively co-invests: it pays 30% of the cost (via immediate deduction) and takes 30% of the return (via the tax), leaving the firm's cost-benefit calculation unchanged.

(e) Under straight-line depreciation, the tax distorts investment (cost of capital > no-tax level), so cutting τ reduces the distortion and increases investment. Under full expensing, the tax is non-distortionary (cost of capital = no-tax level regardless of τ), so cutting τ has no effect on investment. Therefore a rate cut stimulates more investment under straight-line depreciation.

A3: Pollution Externality

(a) Set Demand = PMC: $140 - 2Q = 20 + 2Q \rightarrow 120 = 4Q \rightarrow Q_m = 30$. $P_m = 140 - 2(30) = \$80$.

(b) SMC = PMC + MD $= (20 + 2Q) + Q = 20 + 3Q$. Set Demand = SMC: $140 - 2Q = 20 + 3Q \rightarrow 120 = 5Q \rightarrow Q^* = 24$. $P^* = 140 - 2(24) = \$92$.

(c) DWL $= ½ \times (Q_m - Q^*) \times [SMC(Q_m) - Demand(Q_m)] = ½ \times (30 - 24) \times [(20 + 90) - (140 - 60)] = ½ \times 6 \times (110 - 80) = ½ \times 6 \times 30 = \$90$.

(d) Pigouvian tax $= MD(Q^*) = Q^* = \$24$ per unit. Verify: with tax, firm faces PMC + tax $= 20 + 2Q + 24 = 44 + 2Q$. Set = Demand: $44 + 2Q = 140 - 2Q \rightarrow 4Q = 96 \rightarrow Q = 24 = Q^*$. ✓

(e) If true MD = 3Q (steeper damages), then with a tax set on expected MD = 2Q: $SMC_{expected} = 20 + 4Q$, optimum at $Q = 20$; but true optimum is $SMC_{true} = 20 + 5Q \rightarrow Q = 17.1$. The tax allows Q = 20, which is too much pollution. With permits set at $Q^* = 24$ (based on expected MD), we still overproduce but the quantity is fixed. However, if we had set permits based on expected MD = 2Q at Q = 20, permits perform better when damages are steep because they cap quantity. Weitzman: when the MD curve is steep, getting quantity wrong is very costly, so permits (which fix Q) dominate taxes (which fix price).

A4: Taxation of Savings

(a) Budget constraint: $C_1 + C_2/(1+r) = W \rightarrow C_1 + C_2/1.1 = 100{,}000$. With $U = \ln(C_1) + \ln(C_2)$, the FOC gives $C_2 = (1+r)C_1 = 1.1C_1$. Substituting: $C_1 + 1.1C_1/1.1 = C_1 + C_1 = 2C_1 = 100{,}000 \rightarrow C_1 = \$50{,}000$. $C_2 = 1.1 \times 50{,}000 = \$55{,}000$. $S = W - C_1 = \$50{,}000$.

(b) After-tax rate = 0.05. Budget constraint: $C_1 + C_2/1.05 = 100{,}000$. FOC: $C_2 = 1.05C_1$. Substituting: $C_1 + 1.05C_1/1.05 = 2C_1 = 100{,}000 \rightarrow C_1 = \$50{,}000$. $C_2 = 1.05 \times 50{,}000 = \$52{,}500$. S = $50,000.

(c) Savings unchanged at $50,000. With log utility, substitution and income effects exactly cancel. Substitution effect: lower return makes saving less attractive (save less). Income effect: lower return means you need to save more to reach the same retirement consumption. With U = ln, these exactly offset. C_2 falls from $55K to $52.5K — the tax reduces retirement consumption but not savings.

(d) After-tax wage = 100,000(1-0.20) = $80,000. Budget constraint: $C_1 + C_2/1.1 = 80{,}000$. FOC: $C_2 = 1.1C_1$. Substituting: $2C_1 = 80{,}000 \rightarrow C_1 = \$40{,}000$. $C_2 = \$44{,}000$. S = $40,000. Tax revenue from labor tax = $20,000 in period 1. For part (b), tax revenue = $t \cdot r \cdot S = 0.50 \times 0.10 \times 50{,}000 = \$2{,}500$ collected in period 2. PV = 2,500/1.10 = $2,273. The labor income tax raises far more revenue ($20,000 vs. $2,273) but also distorts more (reduces C_1 and C_2). The two taxes are NOT equivalent at these rates — they would only be equivalent if designed to raise the same PV of revenue.